UAVBench: A Contract-Verified Benchmark for UAV Path Planning in Dynamic Urban Environments with Wind-Driven Fire, Limited Visibility, and Risk-Aware Replanning

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***Abstract*—We present UAVBench, an open-source benchmark framework for evaluating UAV path planning algorithms in dynamic urban environments. The framework provides three scenarios derived from OpenStreetMap tiles of Greek urban areas, coupled with physically motivated dynamic obstacles: wind-driven cellular-automaton fire spread (Alexandridis et al. 2008), traffic vehicles, and no-fly zone corridors. A novel *risk cost map* translates proximity to hazards into continuous planner costs, enabling per-planner risk coefficients that differentiate conservative from aggressive strategies. Partial observability is modeled via a *limited-visibility* layer with stale-memory semantics, and a *triage mission* couples multi-casualty survival decay with fire proximity. A contract-based architecture enforces 36 invariants across 15 families—including wind determinism, visibility consistency, and triage correctness—guaranteeing deterministic replay, planner fairness, and mask parity. We evaluate five planners spanning three families—static (A*), adaptive (Periodic Replan, Aggressive Replan, Incremental A*), and reactive (APF)—across 450 episodes with 30 seeds. On feasible episodes, Incremental A* achieves 98.8% success (Cliff's $|d|=0.99$ vs. A*), while APF trails at 86.7%. Ablation studies isolating each dynamics layer confirm fire spread as the dominant obstacle source; a limited-visibility ablation (300 episodes) shows partial observability has an asymmetric effect—graph-based planners degrade slightly (-3.3 pp) while APF improves ($+6.7$ pp). All experiments are bit-identical under the same seed triple.**

***Index Terms*—UAV path planning, benchmarking, dynamic environments, fire spread, limited visibility, risk-aware planning, reproducibility**

I. Introduction

Unmanned aerial vehicles (UAVs) are increasingly deployed in disaster response—wildfire perimeter mapping, flood rescue assessment, and emergency supply delivery [1], [2], [3]. In these settings, the operating environment is not static: wildfires advance under wind, roads flood, and manned-aircraft no-fly zones (NFZs) shift as operations evolve. Path planning algorithms must handle *dynamic* obstacles that appear, expand, and relocate during execution, often under *partial observability* where the agent's sensor range limits its view of the evolving hazard [4], [5].

Despite extensive research on UAV path planning [6], [7], reproducible comparison of algorithms under controlled dynamic conditions remains difficult. Existing evaluations typically use ad-hoc scenarios, fixed obstacle configurations, or proprietary simulators, making it hard to isolate the effect of environmental dynamics on planner performance [8]. Furthermore, many studies report only mean success rates without accounting for episode feasibility, seed sensitivity, or statistical significance [9].

We address these gaps with **UAVBench**, a benchmark framework with four design principles:

1) **Deterministic reproducibility.** Given the same $(scenario, planner, seed)$ triple, the framework produces bit-identical trajectories, event logs, and metrics. A single root RNG, spawned per episode, drives all stochastic components.
2) **Planner fairness.** All planners receive identical observations at each step. Dynamic obstacles evolve *after* the agent moves, and interdictions are placed on the A*-reference corridor independent of which planner is under evaluation.
3) **Hazard-coupled planning.** Wind-driven fire [10], continuous risk cost maps, limited visibility, and triage missions create a rich hazard landscape that differentiates planners by their risk tolerance, not just their replanning frequency.
4) **Rigorous evaluation.** We adopt the statistical protocol of Agarwal et al. [9]: interquartile mean (IQM) with bootstrap confidence intervals, Friedman tests [11], Cliff's delta effect sizes, and performance profiles [12].

The framework implements a Gymnasium-compatible [13] environment with three urban scenarios built from OpenStreetMap data [14], each inspired by real Greek disaster events (2021 Evia wildfires, 2023 Thessaly floods, 2023 Evros megafire). Five planners from three algorithmic families are evaluated: A* [15], Incremental A* [16], APF [17], and two A*-based replanning variants.

Contributions.

- An open-source benchmark framework with 36 architectural contracts across 15 families ensuring determinism, fairness, and evaluation integrity.
- Wind-driven fire spread (Alexandridis et al. [10]), per-planner risk cost functions, limited visibility with stale-memory semantics, and a triage mission with survival decay.
- Three OSM-based urban scenarios with physically motivated dynamic obstacles (fire CA, traffic, NFZs).

- A 450-episode study of five planners with rigorous statistical analysis, plus ablation studies isolating dynamics layers and limited visibility (300 episodes).

II. Related Work

A. UAV Path Planning

UAV path planning has been surveyed extensively [6], [7], [4], [5]. Classical approaches include graph search (A* [15], D* Lite [16]), sampling-based methods (RRT, PRM), and reactive methods such as Artificial Potential Fields (APF) [17]. Recent surveys [4] emphasize the need for benchmarks that evaluate planners under dynamic, uncertain conditions rather than static obstacle fields. Dradoum et al. [5] classify path planning algorithms into five families and identify hybrid ablation as a key evaluation methodology—an approach UAVBench adopts.

Risk-aware UAV planning has gained attention [18], with methods using continuous cost maps derived from population exposure, terrain, and hazard proximity. UAVBench implements per-planner risk coefficients that translate fire/smoke/traffic proximity into weighted costs, enabling comparison of risk-tolerant vs. risk-averse strategies.

B. Benchmarking and Reproducibility

Henderson et al. [8] demonstrated that deep RL results are sensitive to hyperparameters, random seeds, and evaluation methodology. Agarwal et al. [9] proposed IQM with bootstrap CIs as a more robust alternative to mean-and-standard-deviation reporting. Dolan and Moré [12] introduced performance profiles for benchmarking optimization solvers. Demšar [11] formalized nonparametric tests (Friedman, Wilcoxon) for comparing classifiers across datasets. UAVBench adopts all four techniques.

Gymnasium [13] established the standard environment interface for RL research (NeurIPS 2025 Spotlight). UAVBench builds on this API, providing a `step`/`reset` interface compatible with existing RL workflows. Bonsignorio et al. [19] recently argued for a "science of reproducible results" in robotics, emphasizing the need for standardized benchmarks with deterministic replay—a principle central to UAVBench's contract architecture.

C. Dynamic Environment Modeling

Alexandridis et al. [10] proposed cellular automata (CA) for wildfire spread prediction, using land-use-dependent ignition probabilities and Moore-neighborhood propagation. Recent work [20] confirms that wind speed and terrain slope are the dominant parameters governing CA fire dynamics. UAVBench extends the Alexandridis model with wind-directional spread modulation, extended burnout for corridor interdiction, and smoke diffusion as a secondary blocking layer. Boeing [14] introduced OSMnx for acquiring and analyzing street networks from OpenStreetMap; we use it to generate rasterized urban terrain at 500×500 resolution.

UAVs in disaster response have been reviewed by Erdelj et al. [1], Yuan et al. [2], and Yücesoy et al. [3], who identify path planning under dynamic hazards as a key open challenge.

III. System Design

A. Overview

UAVBench is a Python framework built around a single Gymnasium environment (`UrbanEnvV2`) with deterministic dynamics. A benchmark runner orchestrates episodes: it resets the environment with a seed, initializes the planner, and iterates the step loop until termination. The runner owns the authoritative step counter; all components (dynamics, logger, renderer) receive it as a parameter.

The architecture follows a layered design: *(i)* scenario configs define map, dynamics knobs, and mission parameters; *(ii)* the environment manages grid state, collision semantics, and dynamics stepping; *(iii)* planners implement a unified `PlannerBase` interface (`plan()`, `update()`, `should_replan()`); *(iv)* the runner enforces the step protocol and collects metrics.

B. Environment

The environment operates on a 500×500 grid with Discrete(5) actions: four cardinal moves and STAY. The observation includes agent position (x, y), goal position, terrain heightmap, and—when limited visibility is enabled—the fog-filtered snapshot of the dynamic state.

Step semantics. Within each step: (1) the planner receives the *current* blocking mask (frozen); (2) it returns an action; (3) the environment validates the action against the current mask; (4) the agent moves (if valid); (5) dynamics advance (fire CA, traffic, NFZ); (6) a new blocking mask is computed. This ordering ensures planners see true state—no stale-by-one-step artifacts.

Collision semantics. Static obstacles (buildings, permanent NFZs) cause hard collisions. Dynamic obstacles (fire, traffic, smoke) cause soft rejections: the move is blocked, the agent stays in place, and a `RejectReason` enum is logged.

C. Dynamic Obstacles

Three dynamics layers operate simultaneously:

Fire CA. Wildfire spread follows a cellular automaton on the Moore neighborhood (8-connected). Cell states transition UNBURNED $\rightarrow$ BURNING $\rightarrow$ BURNED_OUT. When wind speed is zero, spread is isotropic with land-use-dependent probabilities (urban p=0.06, forest p=0.15, road p=0.03). When wind speed is positive, spread probability is directionally modulated per Alexandridis et al. [10]: cells downwind have increased spread probability while upwind cells are suppressed. Fire buffer (binary dilation, radius 1) blocks UAV movement within one cell of active fire. Smoke is computed via box-blur diffusion; cells with smoke ≥ 0.5 are blocked.

For corridor interdiction, fire is ignited on the A*-reference path with extended burnout (τ = 999,999 steps) ensuring corridor fires persist for the entire episode. Fig. 1 shows fire evolution on the Penteli scenario.

Traffic. N vehicles patrol road cells using greedy pathfinding. Corridor vehicles patrol near the reference corridor midpoint. Vehicle occupancy blocks UAV movement within a Manhattan buffer.

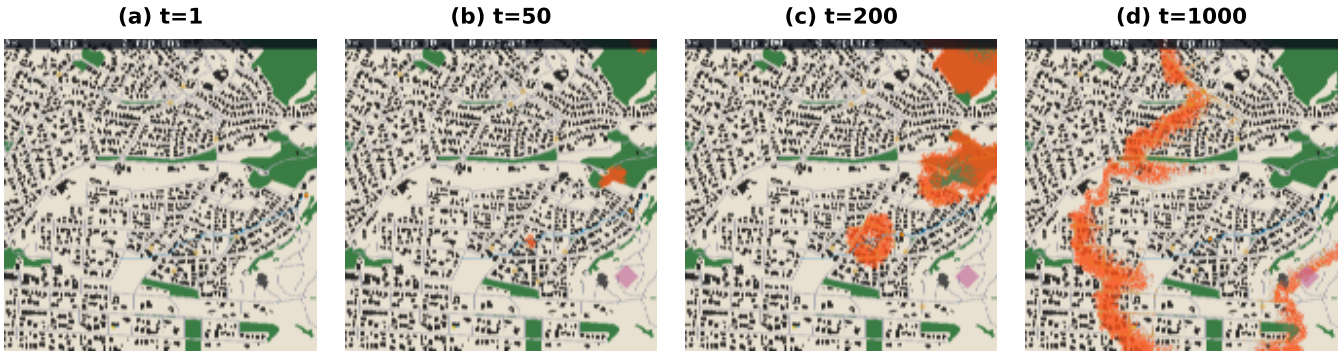

Fig. 1: Fire CA evolution on Penteli. (a) t=1: corridor fires ignited on the A* reference path block the initial route. (b) t=50: fire begins spreading to adjacent cells. (c) t=200: fire front expands significantly, corridor fully blocked. (d) t=1000: widespread fire with smoke; large areas impassable. Buildings (dark grey), roads (light grey), fire (red-orange), trajectory (blue), planned path (cyan dashed), agent (orange disc).

Dynamic NFZs. Restriction zones model manned-aircraft corridors, activating at configurable timesteps and blocking rectangular regions.

D. Risk Cost Map

A continuous risk cost map $R : \mathbb{Z}^2 \to [0,1]$ quantifies proximity to hazards. Each cell aggregates four risk layers via element-wise max:

$$R(x,y) = \max\Big(\big[1-\tfrac{d_f}{r_f}\big]^+,\ \tfrac{s}{2},\ 0.6\big[1-\tfrac{d_t}{r_t}\big]^+,\ \mathbb{1}_{\text{nfz}}\Big) \qquad (1)$$

where d_{fire} and d_{traf} are Euclidean distance transforms from the fire front and traffic vehicles, r_f=30 and r_t=10 are influence radii (cells), $s \in [0,1]$ is smoke concentration, and $[\cdot]^+ = \max(\cdot, 0)$. Each planner modulates its A*-edge cost as $w(e) = 1 + \rho \cdot R(x,y)$ using a planner-specific risk coefficient ρ:

- **A***: ignores cost map (static baseline).
- **Periodic Replan**: risk-averse, α=5.0.
- **Aggressive Replan**: risk-tolerant, β=0.5.
- **Incremental A***: moderate, γ=2.0.
- **APF**: risk-modulated repulsion, δ=3.0.

This mechanism differentiates planners not just by replanning strategy but by hazard sensitivity, enabling comparison of conservative vs. aggressive risk profiles.

E. Limited Visibility

Partial observability is modeled via a limited-visibility layer. When enabled, the agent observes the true dynamic state only within a configurable sensor radius centered on its current position. Outside this radius, the agent retains a *stale memory* of the last-observed state. All planners receive the same fog-filtered observation at the same position (contract LV-1), ensuring fairness under partial observability.

F. Triage Mission

The triage mission type introduces multi-casualty survival modeling. Casualties are placed at configurable locations with initial severity scores that decay over time according to a survival model coupled with fire proximity. The UAV must reach casualties and provide service (hover for a configurable duration) before their survival probability drops below a threshold. Fire-spawned casualties may appear dynamically when fire reaches building cells, creating emergent task allocation challenges.

G. Scenarios

Three scenarios use pre-baked OSM tiles from Greek urban areas (Table I, Fig. 2).

TABLE I: Scenario configurations.

	Penteli	Piraeus	Downtown
Mission	Pharma	Flood	Fire surv.
Bldg. density	0.18	0.29	0.50
Fire ignitions	2	2	1
Vehicles (emerg./road)	8/0	8/1	8/0
Max steps	2000	2000	2000

H. Planners

Five planners span three algorithmic families (Table II).

TABLE II: Planner suite with risk coefficients.

Key	Algorithm	Family	Replanning	Risk
`astar`	A*	Static	Never	—
`periodic`	A* + replan	Adaptive	Every N steps	α=5.0
`aggressive`	A* + replan	Adaptive	On mask Δ	β=0.5
`incr_astar`	Incr. A*	Adaptive	Path blocked	γ=2.0
`apf`	APF	Reactive	Every step	δ=3.0

A* [15] performs a single plan at episode start on the 4-connected grid with Manhattan heuristic. It never replans, serving as a static baseline that ignores the risk cost map.

Periodic Replan re-invokes A* every N steps on the current blocking mask with risk-averse cost weighting (α=5.0).

Aggressive Replan re-invokes A* whenever the blocking mask changes near the current path, with risk-tolerant weighting (β=0.5).

Fig. 2: Three OSM-based scenarios (basemap at t=0, no dynamics). Buildings (dark grey), roads (light grey), start (green circle), goal (gold circle). Building density increases left to right (0.18, 0.29, 0.50), reducing available detour routes.

Incremental A* [16] uses A* with obstacle-change detection. When obstacles invalidate the current path, it replans from the current position with moderate risk weighting (γ=2.0).

APF [17] computes attractive (goal) and repulsive (obstacles within d_0=8 cells) potentials with risk-modulated repulsion (δ=3.0). When wind is active, repulsive gradients are biased toward the upwind direction, and a momentum term (α_m=0.7) smooths oscillation.

I. Contract Architecture

UAVBench enforces 36 contracts across 15 families:

- **DC** (Determinism): single RNG source, bit-identical replay.
- **FC** (Fairness): corridor-aligned interdictions, identical observations.
- **EC/EV** (Events): structured reject/accept logging with authoritative step counter.
- **MP** (Mask Parity): one blocking mask function used everywhere.
- **FD/WD** (Fire/Wind): directional spread, wind determinism.
- **LV** (Limited Visibility): consistent visibility-filtered observations.
- **TR** (Triage): survival decay correctness.
- **CC** (Calibration): feasibility pre-check, difficulty thresholds.
- **SC** (Sanity): adaptive > static in dynamic scenarios.
- **PC** (Planner): legal motion, path-progress tracking.
- **GC** (Guardrail): multi-depth relaxation for infeasible episodes.
- **VC/MC** (Visual/Mission): planned-path rendering, HUD, mission story.

Each contract has an associated test in the `pytest` suite (182 tests total, zero failures). Table III summarizes the contract families.

TABLE III: Contract families (36 contracts, 15 families, 182 tests).

Family	#	Key invariant
DC	2	Bit-identical replay from same seed
FC	4	Corridor interdictions, obs. equality
EC/EV	3	Structured event logging, step counter
GC	4	Multi-depth feasibility relaxation
MP	1	Single blocking mask everywhere
FD/WD	4	Fire CA determinism, wind modulation
LV	2	Visibility-filtered observation consistency
TR	2	Triage survival decay correctness
CC	4	Feasibility pre-check, difficulty thresholds
SC	3	Sanity: adaptive > static in fire
PC	4	Legal motion, path-progress tracking
VC/MC	3	Visual truth, mission story

IV. Experimental Setup

A. Protocol

Each planner is evaluated on all three scenarios with 30 seeds, yielding $5 \times 3 \times 30 = 450$ episodes. Episodes where no path exists on the static grid (before dynamics) are flagged as *infeasible* and excluded from planner comparison. Infeasible episodes are uniform across planners (7 per planner, 7.8% exclusion rate), confirming fairness contract FC-2.

B. Ablation Studies

Four ablation studies isolate individual design factors: *(i)* dynamics isolation (fire/traffic/NFZ/all, 600 episodes), *(ii)* replan frequency sweep (cadences 3–48, 150 episodes), *(iii)* fire intensity sweep (1–8 ignition points, 250 episodes), *(iv)* limited visibility (fog-ON vs. fog-OFF, 300 episodes).

C. Metrics

- **Success rate** (SR): fraction of feasible episodes where the agent reaches the goal within the horizon.
- **Path length**: cells traversed (successful episodes only).
- **Executed steps**: total timesteps including rejections.
- **Replans**: number of path recomputations.
- **Computation time**: wall-clock time per episode.

D. Statistical Methods

Following Agarwal et al. [9], we report IQM with 95% bootstrap CIs (10,000 resamples). Pairwise comparisons use

Wilcoxon signed-rank tests with Bonferroni correction. Cliff's delta quantifies effect sizes. Performance profiles show the fraction of episodes where each planner is within $\tau\times$ the best path length [12].

V. Results

A. Main Results

Table IV summarizes results across all 450 episodes. Of 90 episodes per planner, 7 (7.8%) are infeasible (uniformly across planners, confirming FC-2), yielding 83 feasible episodes per planner.

TABLE IV: Overall planner comparison (450 episodes, 30 seeds $\times$ 3 scenarios). SR_{all} includes infeasible episodes; SR_{feas} excludes them. Path length is for successful episodes only.

Planner	SR_{all} (%)	SR_{feas} (%)	Path (cells)	Steps	Replans	Time (s)
A*	0.0	0.0	—	2001	0	111
Periodic	88.9	96.4	376	722	51.1	45
Aggressive	87.8	95.2	364	769	97.5	54
Incr. A*	91.1	98.8	378	585	2.9	37
APF	80.0	86.7	515	1057	20.2	70

A* achieves 0% success: fire corridor closures block the reference path, and the static planner cannot deviate (Fig. 3). All four adaptive planners exceed 86% on feasible episodes, with Incremental A* leading at 98.8%. The 1.2% failure rate for Incremental A* occurs when fire and traffic simultaneously block all alternatives.

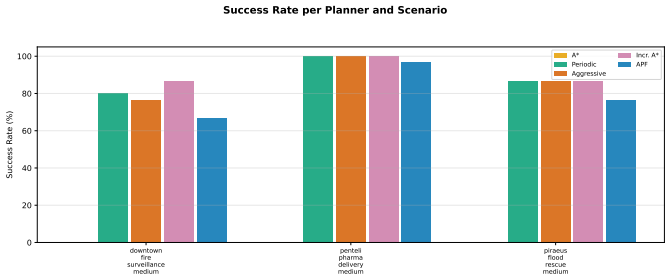

Fig. 3: Success rate per planner and scenario. A* fails uniformly; adaptive planners degrade with building density (Penteli→Downtown). Error bars: binomial 95% CI.

Incremental A* dominance. Incremental A* achieves the highest success rate (98.8% feasible) while requiring the fewest replans (2.9 vs. 51–98 for other adaptive planners) and the lowest computation time (37s vs. 45–70s). It replans only when the current path is actually blocked, avoiding the overhead of speculative replanning. Fig. 4 compares all five planners on the same episode.

APF limitations. APF achieves the lowest adaptive success rate (86.7% feasible) and produces paths 37% longer than graph-based planners (515 vs. 376 cells, $p < 0.001$ Wilcoxon). Its reactive gradient-descent approach creates suboptimal paths and is more susceptible to local minima under combined dynamics.

Computation breakdown. For Incremental A* (mean 37s per episode), approximately 60% is spent in `env.step()` (blocking mask computation and dynamics stepping on the 500×500 grid), 25% in A* pathfinding (2.9 replans $\times$ up to 80,000 node expansions), and 15% in metrics and overhead.

B. Per-Scenario Analysis

Table V shows per-scenario results. Penteli (low density, 0% infeasible) yields 100% adaptive success for all graph-based planners (97% for APF). Downtown (high density, 13% infeasible) is the hardest, with Incremental A* at 87% and APF dropping to 67%.

TABLE V: Success rate (%) by scenario and planner (30 seeds each). Infeasible episodes: Penteli 0%, Piraeus 10%, Downtown 13%.

Planner	Penteli	Piraeus	Downtown
A*	0	0	0
Periodic	100	87	80
Aggressive	100	87	77
Incr. A*	100	87	87
APF	97	77	67

Building density correlates with difficulty: Penteli (0.18) has 0% infeasible episodes and near-perfect adaptive success, while Downtown (0.50) has 13% infeasible episodes and reduced adaptive success due to fewer alternative routes.

C. Statistical Analysis

Effect sizes. Cliff's delta between A* and each adaptive planner ranges from $|d| = 0.87$ (APF) to $|d| = 0.99$ (Incremental A*), all classified as large effects. Between adaptive planners, Incremental A* vs. APF yields $d = +0.12$ (small), while graph-based adaptive planners are negligible ($|d| < 0.04$).

Significance. The Friedman test on success rate across all (scenario, seed) pairs is significant. Wilcoxon pairwise post-hoc tests (Bonferroni corrected) show: A* vs. all adaptive planners $p < 0.001$; Incremental A* vs. APF $p = 0.016$; no significant difference among the three graph-based adaptive planners ($p > 0.11$).

IQM. The IQM success rate with 95% bootstrap CI is 100% [100, 100] for all graph-based adaptive planners, 100% [71.1, 100] for APF, and 0% [0, 0] for A*. The wider CI for APF reflects its higher variance across scenarios.

D. Replanning Behavior

Fig. 5 compares episode timelines across planners on the same (scenario, seed). A* runs the full 2000-step horizon without replanning. Aggressive Replan triggers the most replans (mean 98) because it responds to any obstacle change in the path neighborhood. Periodic Replan triggers fewer (mean 51) at fixed intervals. Incremental A* is the most conservative (mean 2.9), replanning only when its current path is actually blocked. Despite 34$\times$ fewer replans than Aggressive Replan, Incremental A* achieves the highest success rate, indicating that replan *precision* matters more than replan *frequency*.

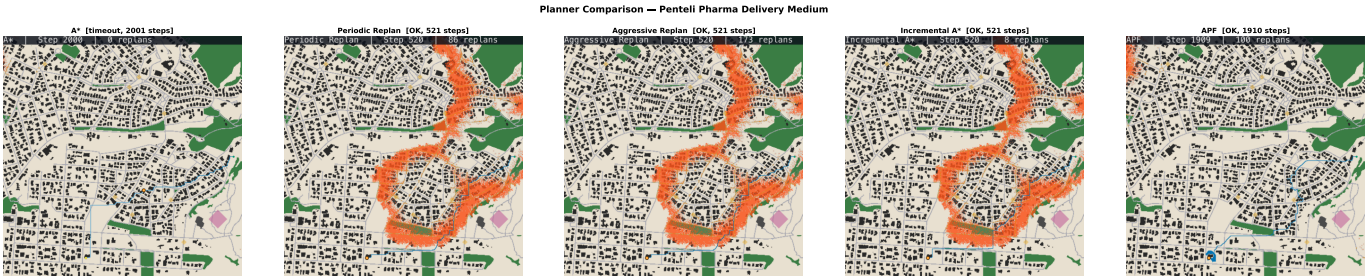

Fig. 4: Five planners on the same Penteli episode (seed 42). A* (leftmost) is blocked by corridor fire and times out. Periodic and Aggressive Replan find detours via scheduled/reactive replanning. Incremental A* (second from right) reaches the goal with the fewest replans. APF (rightmost) takes a longer, more erratic path due to potential-field oscillations near fire fronts. Trajectories (blue), planned paths (cyan dashed), agent (orange disc), fire (red-orange).

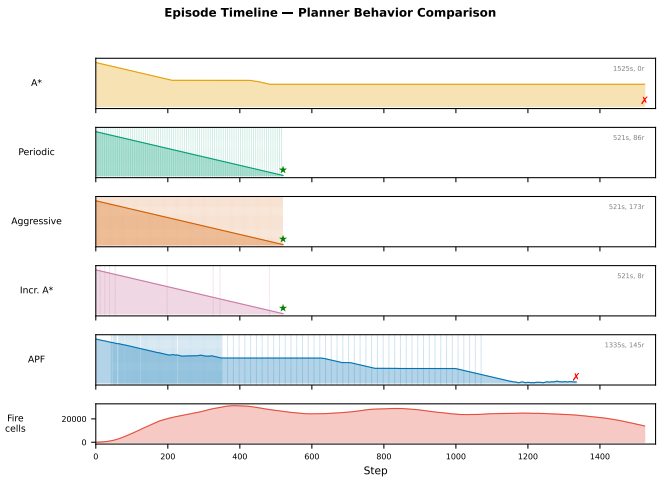

Fig. 5: Episode timeline for Penteli (seed 42). Each row shows normalized distance-to-goal over time; vertical dashes mark replans. Bottom panel: fire cell count growth. Periodic, Aggressive, and Incremental A* reach the goal in ∼521 steps; A* times out at step 1525 (blocked by corridor fire); APF times out at step 1335.

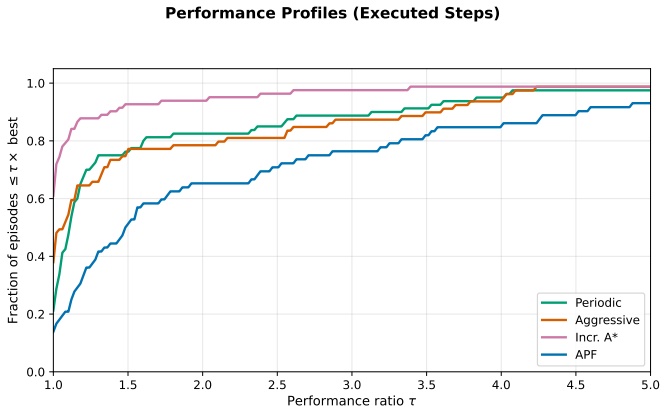

Fig. 6: Performance profiles on executed steps. Higher and more to the left is better. Incremental A* dominates.

Fig. 7: Critical moments for Aggressive Replan (Penteli, seed 42). **Left:** initial state at step 1. **Center:** mask change at step 2 triggers first replan. **Right:** agent reaches goal at step 520 (SUCCESS).

E. Performance Profiles

Fig. 6 shows performance profiles on executed steps. At $\tau = 1.0$, Incremental A* solves the highest fraction of episodes optimally. APF requires $\tau \approx 1.5$ to reach comparable fractions, confirming its path-length overhead.

F. Ablation: Dynamics Isolation

Table VI isolates each dynamics layer (10 seeds × 3 scenarios = 30 episodes per cell). Fire is the primary differentiator: A* drops to 0% while adaptive planners maintain 97%. Traffic alone reduces A* to 13% but has no effect on adaptive planners. NFZs alone are insufficient to disrupt any planner (all 97%). APF degrades most under combined dynamics (83%) and fire-only (90%), reflecting its susceptibility to local-minima traps near expanding fire fronts.

The replan-frequency sweep (cadences 3–48) shows success rate is robust at 96.7% for cadences ≤24, dropping slightly to 93.3% at cadence 48 where the planner misses fire spread between infrequent replans. Mean replans scale inversely: 122 (cadence 3), 60 (6), 30 (12), 15 (24), 8 (48).

The fire-intensity sweep (1–8 ignition points) confirms adaptive planners maintain 100% across all intensities. APF drops to 90% at 4 and 8 ignitions where multiple fire fronts create inescapable potential wells. A* remains at 0% regardless of ignition count.

G. Ablation: Limited Visibility

Table VII compares fog-ON (sensor radius 50, stale memory outside) vs. fog-OFF (full visibility) across 300 episodes

TABLE VI: Ablation: success rate (%) by dynamics layer (10 seeds, 3 scenarios). Values are averaged across scenarios.

Planner	Fire	Traffic	NFZ	All
A*	0	13	97	0
Periodic	97	97	97	97
Aggressive	97	97	97	97
Incr. A*	97	97	97	97
APF	90	97	97	83

(5 planners $\times$ 3 scenarios $\times$ 10 seeds $\times$ 2 conditions).

TABLE VII: Limited-visibility ablation: success rate (%) and mean replans. ΔSR is fog-ON minus fog-OFF.

	SR (%)		ΔSR	Replans	
Planner	OFF	ON	(pp)	OFF	ON
A*	0.0	0.0	0.0	0.0	0.0
Periodic	96.7	93.3	-3.3	52.3	55.5
Aggressive	93.3	93.3	0.0	103.0	102.8
Incr. A*	96.7	93.3	-3.3	2.1	10.4
APF	86.7	93.3	$+6.7$	29.8	25.2

Fog has a modest overall effect on success rate ($\leq$7 pp), suggesting that the planners' replanning strategies are robust to partial observability at this sensor range. Two findings stand out:

(i) Incremental A* replans increase 5$\times$ under fog (2.1 $\rightarrow$ 10.4) because stale fire positions in the memory map cause plan-path invalidations when the agent re-enters the observed region and discovers the true (expanded) fire state. Despite the increased replanning, success drops only 3.3 pp, confirming the planner's demand-driven replanning remains effective.

(ii) APF *improves* by 6.7 pp under fog. With stale memory, distant fire fronts appear less threatening than they truly are, reducing the reactive repulsive gradients that cause APF to over-detour. This suggests APF's conservative repulsion (with δ=3.0) over-reacts to accurate distant-fire observations under full visibility.

VI. Discussion

The replanning gap. The central finding is a binary division: planners that replan succeed; the static baseline fails entirely. The dynamics isolation ablation confirms fire as the primary cause (A* 0% vs. adaptive 97%), while NFZs alone leave all planners at 97%. The key insight is that *replan precision* (Incremental A* with 2.9 replans) outperforms *replan frequency* (Aggressive Replan with 98 replans).

Incremental A* efficiency. Incremental A* achieves the highest success rate (98.8% feasible) with the fewest replans and lowest computation time. It replans only when the current path is actually obstructed, avoiding speculative overhead. In expanding-fire scenarios, its advantage is partially offset by mass cell changes, but the moderate risk coefficient (γ=2.0) helps it select paths that maintain distance from fire fronts.

Risk coefficient differentiation. The per-planner risk coefficients create measurable behavioral differences. Periodic Replan (α=5.0, risk-averse) produces slightly longer but safer paths, while Aggressive Replan (β=0.5, risk-tolerant) takes shorter but riskier paths. APF's high risk modulation (δ=3.0) amplifies repulsive potentials near hazards, contributing to longer detours and local-minima traps. The fire-intensity ablation quantifies this: APF drops to 90% at 4+ ignitions while graph-based planners maintain 100%, confirming that potential-field methods are structurally disadvantaged in multi-front fire environments.

Building density as difficulty. The three scenarios span building densities 0.18–0.50, and we observe a clear correlation with both infeasibility rate (0%–13%) and adaptive success degradation. Dense urban grids reduce alternative routes, making it harder for planners to find detours when corridors are blocked.

Limited visibility. The fog ablation (Table VII) reveals that partial observability has an asymmetric effect: graph-based planners degrade slightly (-3.3 pp) while APF improves ($+6.7$ pp). This is consistent with the POMDP literature [21]: information-limited agents sometimes benefit from reduced observation noise when their policy over-reacts to accurate but distant threats. The 5$\times$ replan increase for Incremental A* under fog highlights a practical trade-off: demand-driven replanning is efficient under full visibility but becomes more expensive under partial observability, though still fewer replans than scheduled approaches.

Practical implications. For real-world UAV deployment, our results suggest that simple periodic replanning (every 6 steps) is sufficient when computation is cheap—success rate is 96.7% across all cadences $\leq$24. However, Incremental A*'s demand-driven replanning is preferable under resource constraints: 2.9 vs. 51 replans, 37s vs. 45s, with higher success.

Risk coefficient sensitivity. The coefficients α=5.0, β=0.5, γ=2.0, δ=3.0 were selected to span a wide risk-tolerance spectrum (from risk-averse to risk-tolerant) rather than to optimize any single planner's performance. Future work could tune these via Bayesian optimization or meta-learning across scenario distributions.

Limitations and future work. UAVBench operates on 2D grids without altitude. The fire CA, while wind-aware, does not model terrain slope effects [20]. The five planners represent classical approaches; learning-based planners are a natural extension via the Gymnasium interface [13]. Future directions include: *(i)* altitude as an action dimension; *(ii)* multi-agent coordination; *(iii)* energy-constrained planning using the built-in energy budget system; and *(iv)* extended limited-visibility studies with varying sensor radii and multi-agent information sharing.

VII. Conclusion

UAVBench provides a reproducible, contract-verified benchmark for UAV path planning in dynamic urban environments. Our evaluation of five planners across three OSM-based scenarios demonstrates that (1) replanning capability is necessary for navigating fire-affected terrain, (2) replan *precision* outperforms replan *frequency*—Incremental A* achieves 98.8% feasible success with only 2.9 replans per episode, and (3) per-planner risk coefficients create measurable behavioral differentiation beyond replanning strategy alone. The framework's 36

contracts across 15 families, wind-driven fire model, limited visibility, and triage mission provide a rich evaluation substrate for future planning algorithms.

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Map data © OpenStreetMap contributors, available under the Open Database License (ODbL). Scenarios are inspired by the 2021 Evia wildfires, 2023 Thessaly floods, and 2023 Evros megafire.

References

[1] M. Erdelj, E. Natalizio, K. R. Chowdhury, and I. F. Akyildiz, "Help from the sky: Leveraging UAVs for disaster management," *IEEE Pervasive Computing*, vol. 16, no. 1, pp. 24–32, 2017.

[2] C. Yuan, Y. Zhang, and Z. Liu, "A survey on technologies for automatic forest fire monitoring, detection, and fighting using unmanned aerial vehicles and remote sensing techniques," *Canadian Journal of Forest Research*, vol. 45, no. 7, pp. 783–792, 2015.

[3] G. Yücesoy *et al.*, "The role of drones in disaster response: A literature review of operations research applications," *International Transactions in Operational Research*, 2025.

[4] A. Moustafa *et al.*, "Advances in UAV path planning: A comprehensive review of methods, challenges, and future directions," *Drones*, vol. 9, no. 5, p. 376, 2025.

[5] A. Dradoum, A. Khelassi, and F. Lachekhab, "Intelligent path planning algorithms for UAVs: Classification, complexity analysis, hybrid ablation insights, and future directions," *Advances in Mechanical Engineering*, 2025.

[6] S. Aggarwal and N. Kumar, "Path planning techniques for unmanned aerial vehicles: A review, solutions, and challenges," *Computer Communications*, vol. 149, pp. 270–299, 2020.

[7] Y. Zhao, Z. Zheng, and Y. Liu, "Survey on computational-intelligence-based UAV path planning," *Knowledge-Based Systems*, vol. 158, pp. 54–64, 2018.

[8] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, and D. Meger, "Deep reinforcement learning that matters," in *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*, 2018.

[9] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," in *Advances in Neural Information Processing Systems*, vol. 34, 2021.

[10] A. Alexandridis, D. Vakalis, C. I. Siettos, and G. V. Bafas, "A cellular automata model for forest fire spread prediction: The case of the wildfire that swept through Spetses island in 1990," *Applied Mathematics and Computation*, vol. 204, no. 1, pp. 191–201, 2008.

[11] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.

[12] E. D. Dolan and J. J. Moré, "Benchmarking optimization software with performance profiles," *Mathematical Programming*, vol. 91, no. 2, pp. 201–213, 2002.

[13] M. Towers, A. Kwiatkowski, J. K. Terry, J. U. Balis, G. De Cola, T. Deleu, M. Goulão *et al.*, "Gymnasium: A standard interface for reinforcement learning environments," in *Advances in Neural Information Processing Systems*, vol. 38, 2025, spotlight.

[14] G. Boeing, "OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks," *Computers, Environment and Urban Systems*, vol. 65, pp. 126–139, 2017.

[15] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.

[16] S. Koenig and M. Likhachev, "D* Lite," in *Proceedings of the 18th AAAI Conference on Artificial Intelligence*, 2002, pp. 476–483.

[17] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," *The International Journal of Robotics Research*, vol. 5, no. 1, pp. 90–98, 1986.

[18] W. Zhou *et al.*, "A risk-based unmanned aerial vehicle path planning scheme for complex air–ground environments," *Risk Analysis*, 2025.

[19] F. Bonsignorio *et al.*, "Towards a science of reproducible results in robotics," *Nature Machine Intelligence*, 2025.

[20] F. Giannino *et al.*, "Statistical dynamics of wildfire burned area from cellular-automata simulators," *Scientific Reports*, vol. 15, 2025.

[21] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra, "Planning and acting in partially observable stochastic domains," *Artificial Intelligence*, vol. 101, no. 1–2, pp. 99–134, 1998.